BE

1. $V_1 = .005\ m^3$ $\quad$ $p_1 = 5\ bar$

$V_f\ 1.5\ m^3$ $\quad$ $p_2 = ?$

$v_f = 1.00158$

$v_g = .2503$

$m_f\ \frac{V_f}{v_f} = \frac{.005\ m^3}{.00158} = \boxed{\cancel{3.16\ kg}} = .00499\ kg$

$m_g\ \frac{V_g}{v_g} = \frac{1.5}{.2503} = 5.99\ kg$

$5.99\ m = 6.\ kg$

$\alpha = \frac{m_g}{m} \quad \frac{5.99}{6} = .999$

$v\ v_f + \alpha\,(v_g - v_f)$

$v = 1.00158 + .999\,(.2503 - 1.00158)$

$v = .25105\ m^3/kg$

$u_f = u_f + \alpha\,(u_g - u_f)$

$u_1 = 198.39 + .999\,(1321.02 - 198.39)$

$\boxed{u_1 = 1318.39}$

State 2

$v_0 = V_2 = V_g = .225 \, m^3/kg$

$PV =$

bar
45 75
x
.5

b. $Q = m(U_2 - U_1)$
.005 C

2. $m = 18\,kg$
$P_1 = 100\,kPa$
$T_1 = 263\,K$ $n = 1.1$
$P_2 = 550\,kPa$

$$\frac{T_2}{263} = \frac{550}{100}^{\frac{1.1-1}{1.1}}$$

$T_2 = 307\,K$

$W_1 = \dfrac{P_1V_1 - P_2V_2}{n-1} = \boxed{-3373.34\ KJ}$

$k = 1.33$

B. $W\left(\dfrac{k-n}{n-1}\right) = \boxed{-2699.6\ KJ}$

3. $m = 2.8\ kg$

$P_1\ 2\ bar$ ~~[illegible]~~ $\quad k = 1.4$

$T = 493.15\ K$ $\quad V_I =$

$V_f = 2V_I$

A. ~~$v = \frac{V}{m}$ .26~~ $\quad PV = mRT$

~~$2V = .26\ 3.314\ 493.15$~~

~~$491.72\ m^3$~~

$V = \dfrac{mRT}{P} = \boxed{.1689346\ m^3/kg}$

B. $W = P(V_2 - V_1) = \boxed{33.946\ KJ}$

2(

$P_1 = P_2$

$V_1 = 2V_1$

C. T

4. $p_1 = 1$ bar $= 10$ Mpa $V_2 = .003$

$p_2 = 218K$. $n = 1.25$

$V_1 = .25 m^3$

$p_1 V_1^n = p_2 V_2^n$.

$\cancel{m_1 = \frac{p_1 V_1}{R T_1}}$

$p_2 = p_1 \left(\frac{V_1}{V_2}\right)^{1.25} = 14.169$ Mpa

200C

a $W = \frac{p_1 V_1 - p_2 V_2}{1.25 - 1} = \boxed{63.30392 \text{ MJ}}$

b $63.3092 \left(\frac{1.4 - 1.25}{1.4 - 1}\right) = \boxed{25.6128 \text{ KJ}}$